

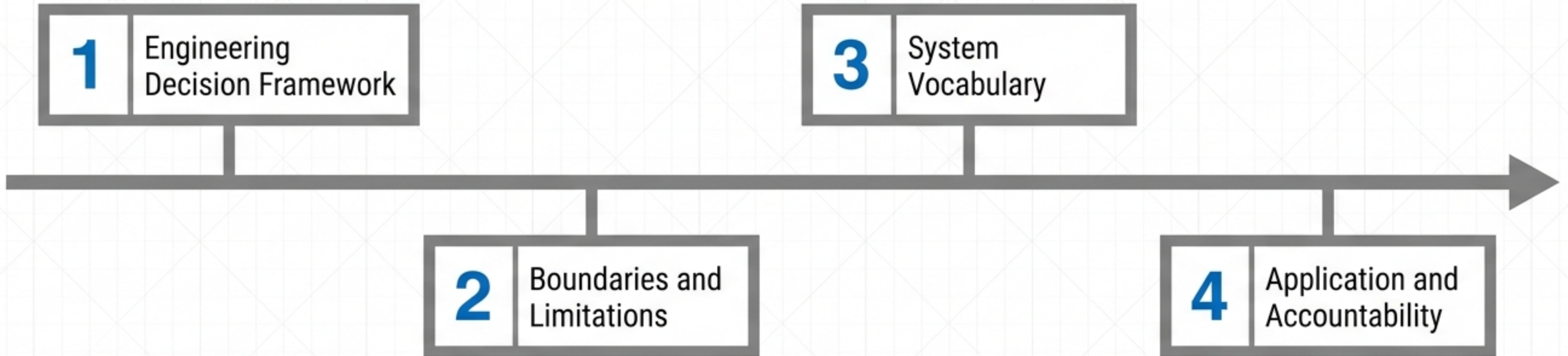


Module 1, Lesson 3

AI 101 FOR MECHANICAL AI 101 FOR MECHANICAL ENGINEERS

Start with an engineering decision

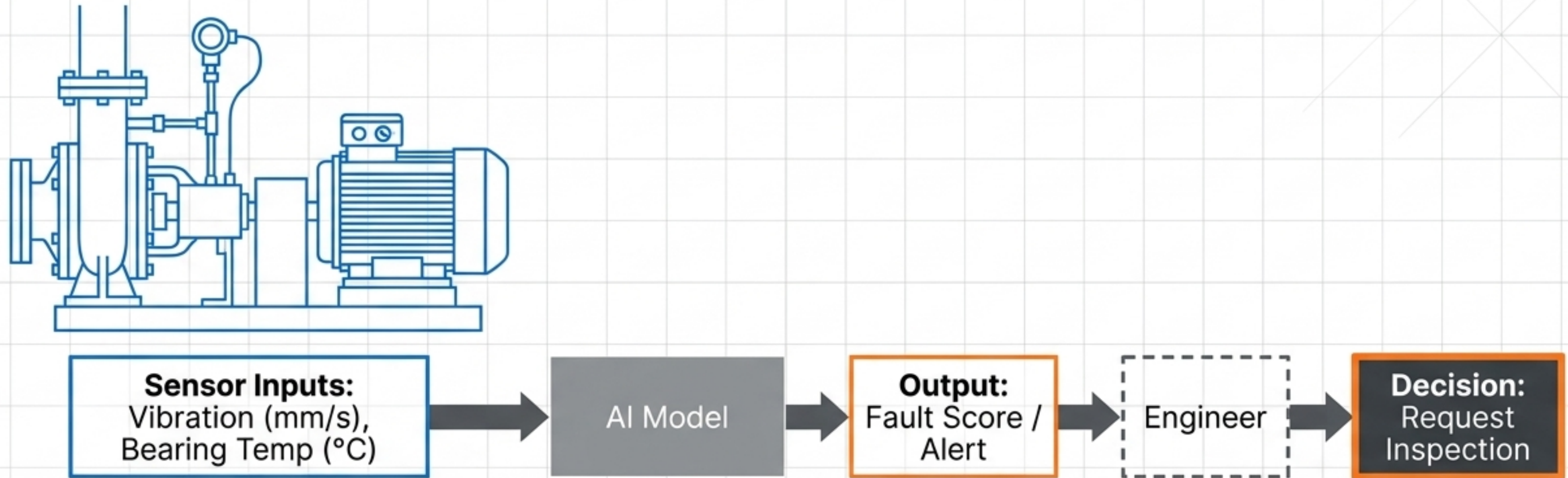
Outline



1

Engineering Decision Framework

Sequence of a Maintenance Decision



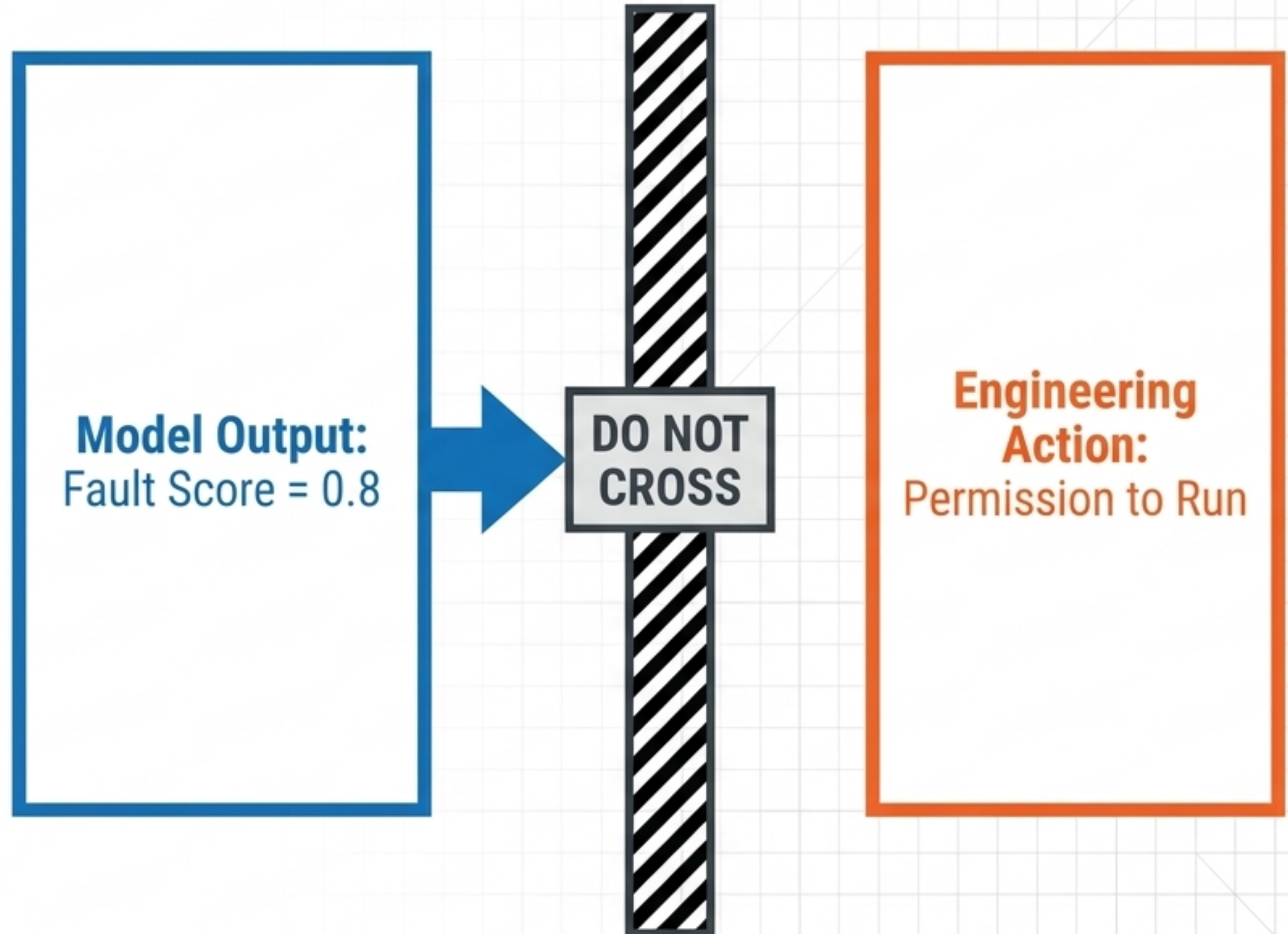
- Start projects with decisions, not algorithms.
- Establish available inputs before deciding.
- Define outputs as specific fault alerts.
- Balance fault detection against false alarms.

2

Boundaries and Limitations

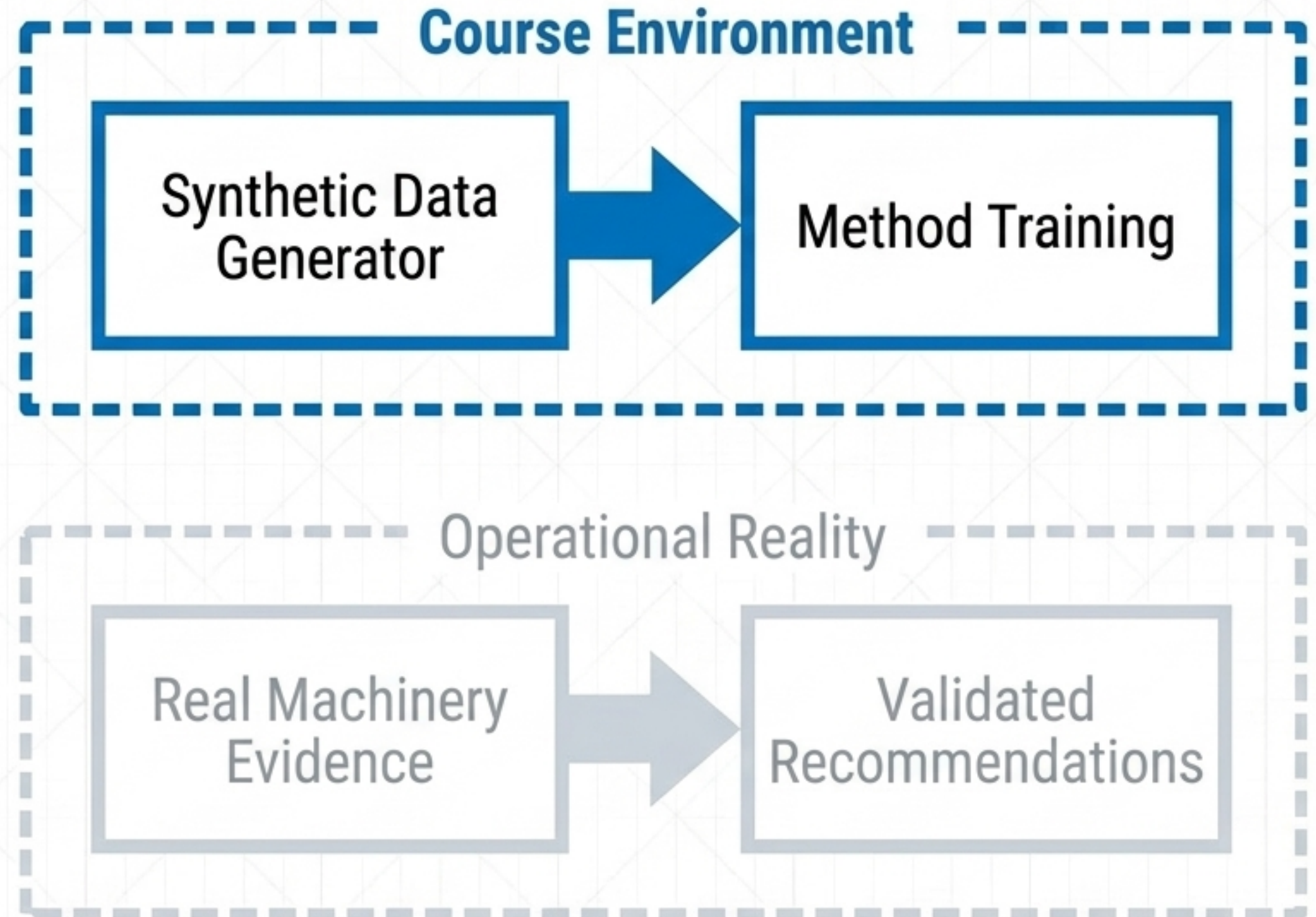
Distinguishing Estimates From Actions

- Model outputs provide statistical estimates only.
- Estimates are not permission to operate.
- Engineers own operating limits entirely.
- Engineers control measurement quality and validation.
- Practical systems require defined escalation paths.



Scope of Synthetic Examples

- Synthetic examples teach analytical methods.
- They avoid validated machinery recommendations.
- Apply synthetic cases to practice reasoning.
- Evaluate evidence and consequences systematically.

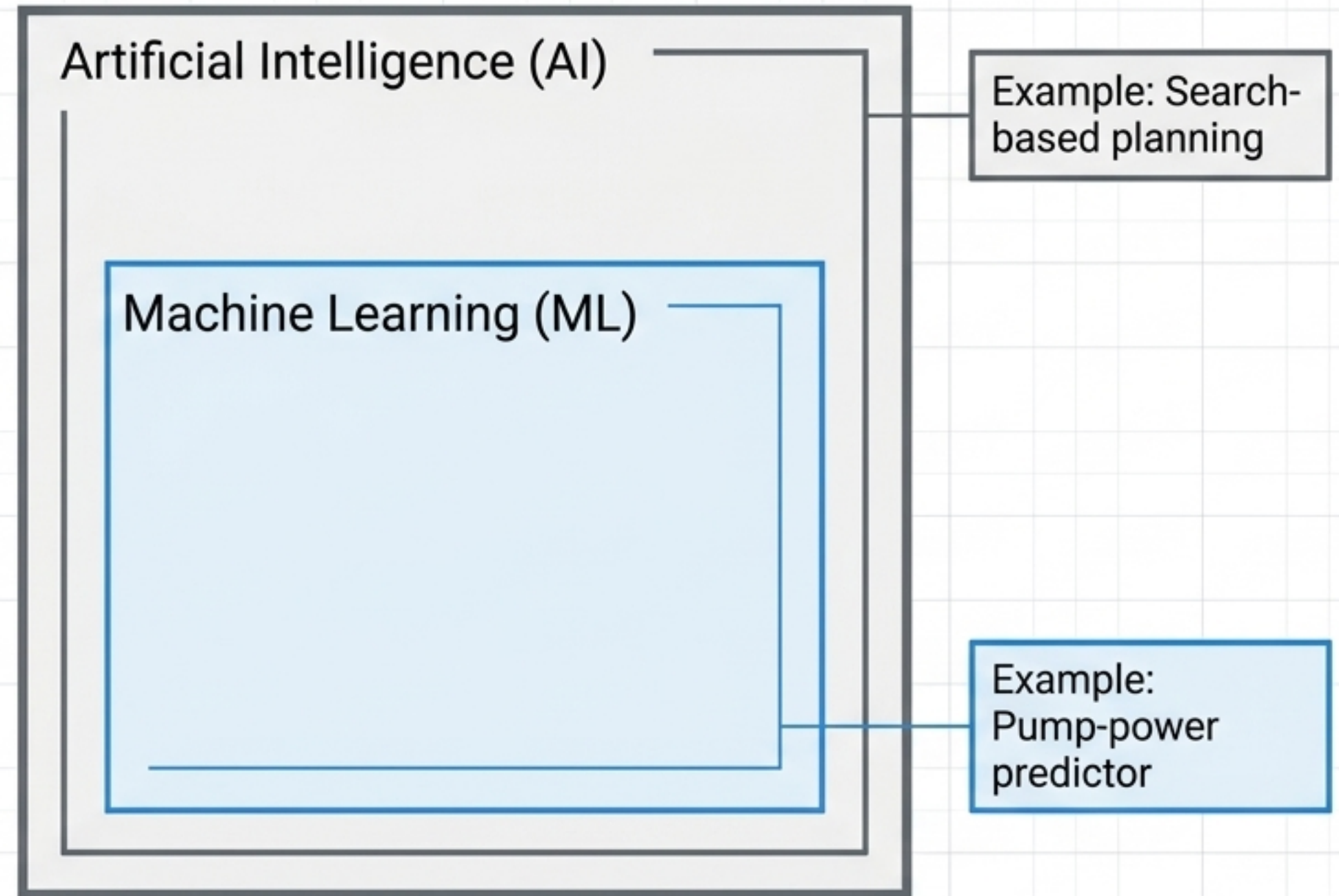


3

System Vocabulary

Hierarchy of Intelligent Systems

- AI encapsulates systems performing intelligent tasks.
- Search-based planning represents one AI approach.
- Machine learning forms a specific subset.
- ML algorithms learn patterns from data.
- Predictors fit from measured operational examples.



4

Application and Accountability

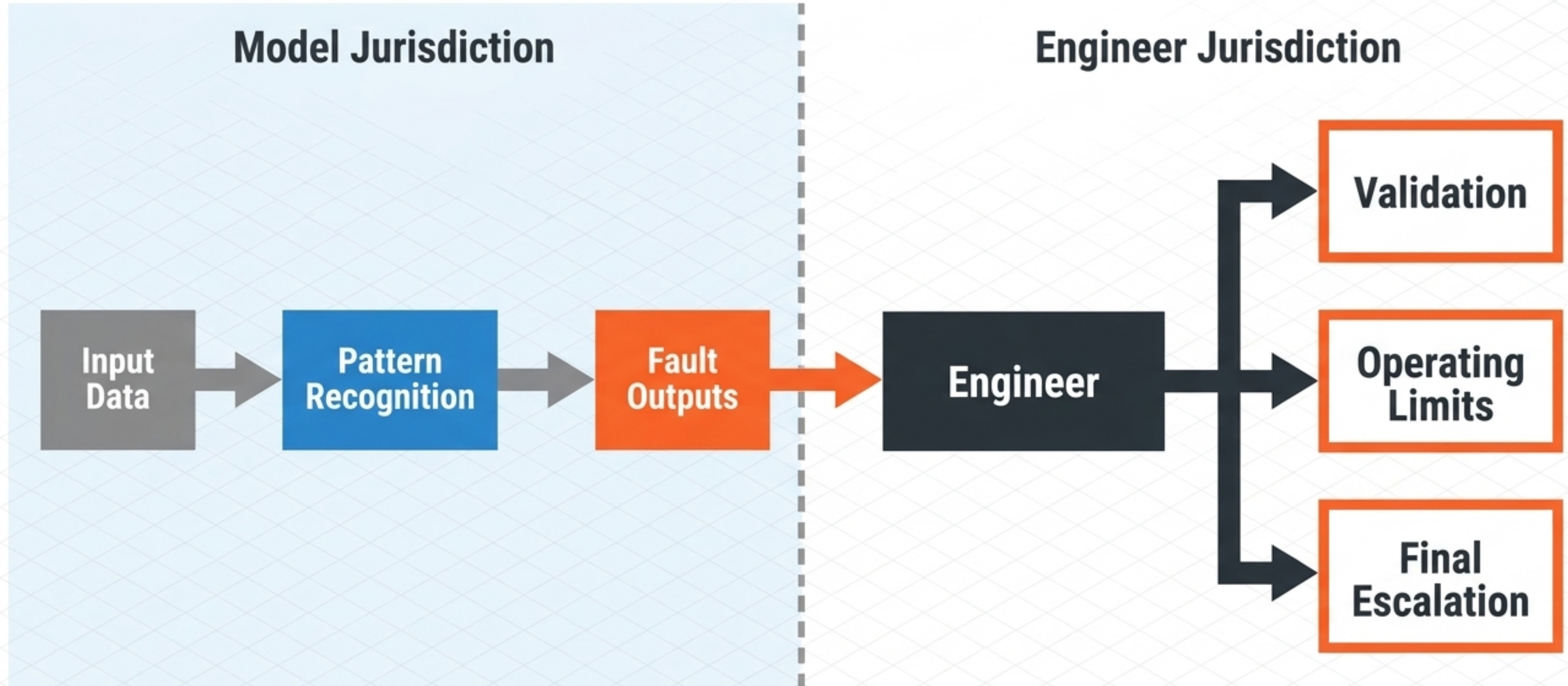
Component Diagnostic Matrix

- Monitoring systems combine multiple functional components.
- Component roles dictate required operational evidence.
- Equipment labels omit underlying learning mechanisms.
- Evaluate learned detectors on held-out conditions.

System Component	Diagnostic Function	Action Prerequisite
Fixed Trip	Implements automation rule via limit	Requires threshold breach
Learned Detector	Predicts fault scores from examples	Requires held-out condition evaluation
Maintenance Assistant	Generates operational text	Requires verification against records

Jurisdictional Accountability Map

- AI tools solely support engineering decisions.
- Decisions give inputs precise meaning.
- Accountability remains with human operators.
- Engineers define limits and escalation.



Key Points

1

Define maintenance decisions using specific inputs and outputs.

2

Separate model estimates from the operational actions they support.

3

Machine learning is a data-driven subset of artificial intelligence.

4

Engineers retain absolute accountability for operating limits and escalation.